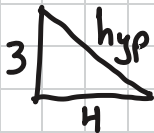


Homework

Show in $z = x + yi$

① $|z| = 5$, $\arg(z) = \tan^{-1} \frac{3}{4}$

$$\text{hyp} = \sqrt{3^2 + 4^2} = 5$$



$$\cos \theta = \frac{4}{5}$$

$$\sin \theta = \frac{3}{5}$$

$$x = r \cos \theta = 5 \left(\frac{4}{5} \right) = 4$$

$$y = r \sin \theta = 5 \left(\frac{3}{5} \right) = 3$$

$$z = 4 + 3i$$

② $|z| = \pi$, $\arg(z) = \frac{\pi}{6}$

$$z = r \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$$

$$= \pi \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i \right)$$

$$= \frac{\pi\sqrt{3}}{2} + \frac{\pi}{2}i$$

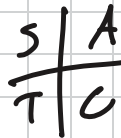


③ $|z| = \frac{1}{2}$, $\arg(z) = -\frac{\pi}{3}$

$$z = r \left(\cos -\frac{\pi}{3} + i \sin -\frac{\pi}{3} \right)$$

$$= \frac{1}{2} \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i \right)$$

$$= \frac{1}{4} - \frac{\sqrt{3}}{4}i$$



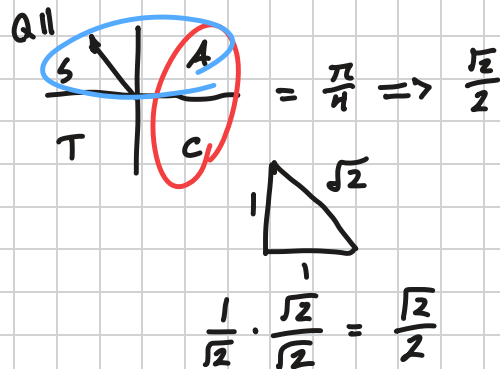
$$\cos\left(-\frac{\pi}{3}\right) = \cos \frac{\pi}{3}$$

$$\sin\left(-\frac{\pi}{3}\right) = -\sin \frac{\pi}{3}$$

④ $|z| = 1, \arg(z) = \frac{3\pi}{4}$

$$\begin{aligned} z &= r \left(\cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right) \right) \\ &= 1 \left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \right) \\ &= -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \end{aligned}$$

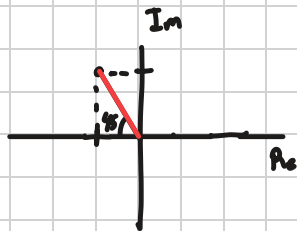
$$\begin{aligned} \cos\left(\frac{3\pi}{4}\right) &= -\frac{\sqrt{2}}{2} \\ \sin\left(\frac{3\pi}{4}\right) &= +\frac{\sqrt{2}}{2} \end{aligned}$$



⑤ $z = -1 + i$

$$|z| = r = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\theta = \tan^{-1}\left(\frac{1}{-1}\right) = \frac{3\pi}{4}$$



⑥ $z = 3i$

$$|z| = r = \sqrt{0^2 + 3^2} = 3$$

$$\theta = \frac{\pi}{2}$$

